Answers

1.**A**	2.**D**	3.**B**	4.**A**	5.**C**	6.**A**	7.**2**	8.**C**
9.**C**	10.**E**	11.**A**	12.**A**	13.**C**	14.**A**	15.**8**	16.**1**
17.**9**	18.**7**	19.**6**	20.**240**	21.**A**	22.**A**	23.**2**	24.**C**
25.**13**	26.**A**	27.**0**	28.**B**	29.**A**	30.**6**	31.**A**	32.**3**
33.**D**	34.**E**	35.**A**	36.**D**	37.**C**	38.**E**	39.**7**	40.**E**
41.**A**	42.**A**	43.**20**	44.**D**	45.**A**	46.**B**	47.**E**	48.**C**
49.**A**	50.**D**	51.**1200**	52.**C**	53.**A**	54.**B**		

Explanations

1.**A**

Amiya runs a campaign on issues, and we need to find the maximum vote share that she can get.

We are trying to maximise the number of voters for Amiya, that means Amiya needs to run a vigorous campaign.
Since, we are trying to increase the vote share for Amiya, we want as many voters as possible transferred from Ramya's share to Amiya's votes.
As the number of votes a candidate receives is proportional to the intensity level of the campaign, we want Ramya also to run a vigorous AND attacking campaign, so that her votes are transferred to Amiya.

In this scenario we have 20x(2+2)% of the people voting, 80% of the people. Of that, if both had ran issues campaign, they would have each received 40% of the votes.

Now, we want Ramya to run an attacking campaign, where 20% of the people that would have voted for her vote for Amiya.
So 20% of 40% of the votes are transferred to Amiya. That is 8% of the votes. And we are also told that, 5% that would have voted for her dont vote, so 5% of 40% dont vote, that is 2% of the voters.

Final Tally is Amiya gets 48% of the votes, and Ramya gets 30% of the votes.

VIDEO SOLUTION

2.**D**

If both Ramya and Amiya run staid campaigns, the intensity of each staid campaign is 1
So total number of voters that would vote for them if they focused on issues will be 20x(1+1)%=40%

This means if they had both ran regarding issues, then they would get 20% of votes each.
We are told that both of them run attacking campaigns,
And the rule for mutual attacking campaign is 10% of voters who would have voted for each candidate will not vote.
That means 10% of 20% of each candidate will not vote, now that it is a mutually attacking campaign.
That means, each candidate receives 18% of the votes.

Total votes received is 36%.

VIDEO SOLUTION

3.**B**

We are looking for the minimum possible number of votes that Ramya can get when she runs an attacking campaign.

To minimise the number of votes, we can have Ramya run a staid campaign to minimise the votes, so minimum intensity, which will get her 20% of the votes if she ran with issues. Now that she is running with attacking, she will loose 20% of the votes to Amiya and 5% of the votes will not vote anymore.

That is a total 25% loss. Remaining votes she will get is 75% of the 20% which will leave her with 15% of the votes.

VIDEO SOLUTION

4. **A**
Let the share prices at 10,11,12,1,2,3 be p,q,r,s,t,u
Since Abdul lost all money, p>u
From Dane it can be observed, s+t+u>p+q+r
p>r and u>t
From Emily it can be observed, r+u>p+s
From these inequalities it can be inferred that the p is the highest.

VIEW SOLUTION

5. **C**
Check out the options.

In last row, only one individual can't be reached by anyone.

In the fourth row, only one individual can't be reached by anyone.

In the middle column, every individual can be reached by atleast another individual.

In the fourth column, both 1 and 2 cannot be reached by any person. Hence, fourth column is the correct answer.

VIDEO SOLUTION

6. **A**
Let us make note of the information given in the set.
The set states that rural kids were surveyed from 3 regions NE, West and South.
50 students each were surveyed from 150 villages in the NE, 250 villages from the West and 200 villages from the South.

Total number of kids from the NE = 150*50 = 7500
Total number of kids from the West = 250*50 = 12500
Total number of kids from the South = 200*50 = 10,000.

The table, given in the question, gives the number of students whose mothers dropped out before completing primary education. Therefore, the mothers of the remaining students should have completed primary education.

There are 7500 students from the NE in total. The mothers of 5000 of those students dropped out before completing primary education. Therefore, the remaining 2500 kids should have mothers who have completed primary education.

Let us make 2 tables - one representing the number of students and the other representing the number of students whose mother has completed primary education.

55% of the surveyed kids studied in schools run by the Government.
Number of kids studying in Govt. School should be 0.55*30000 = 16,500

From the given table, we know that the mothers of 13,500 kids who went to Govt. Schools dropped out of primary school.
Therefore, the remaining 16,500 - 13,500 = 3000 kids should have mother who have completed primary school.

37% of the surveyed kids study in private schools.
Number of kids studying in private schools should be 0.37*30000 = 11,100.
Number of kids in private schools whose mothers have completed primary school = 11,100 - 2700 = 8400.

8% of the surveyed kids did not go to school.
Number of kids who did not go to school = 0.08* 30000 = 2400
Number of kids not going to schools whose mothers have completed primary school = 2400 - 1800 = 600.

In S, 60% of the surveyed kids were in G.
Therefore, 6000 kids in S must be from G.

In NE, among the O kids, 50% had mothers who had dropped out before completing primary education.
There are 300 O kids whose mothers dropped out before completing primary education. These kids represent 50% of the total number of O-kids. Therefore, there must be 600 O-kids from NE - 300 kids should have mothers who dropped out before completing primary education and 300 kids should have mothers who have completed primary education.

The number of kids in G in NE was the same as the number of kids in G in W. Therefore, the number of kids in G in NE and the number of kids in G in W should be equal to 5250. Total number of kids:

	G	P	O	Total
NE	5250	1650	600	7500
W	5250			12500
S	6000			10000
Total	16,500	11,100	2,400	30,000

Number of kids whose mothers have completed primary education:

	G	P	O	Total
NE	1050	1150	300	2500
W	1050			5200
S	900			4300
Total	3000	8400	600	12000

We have been given that in S, all surveyed kids whose mothers had completed primary education were in school. Therefore, the number of kids not going to school whose mothers have completed primary education in S should be 0. Filling the tables, we get,

Total number of kids:

	G	P	O	Total
NE	5250	1650	600	7500
W	5250	5750	1500	12500
S	6000	3700	300	10000
Total	16,500	11,100	2,400	30,000

Number of kids whose mothers have completed primary education:

	G	P	O	Total
NE	1050	1150	300	2500
W	1050	3850	300	5200
S	900	3400	0	4300
Total	3000	8400	600	12000

There are 300 students who are not in school in W whose mothers have completed primary education. Therefore, option A is the right answer.

VIDEO SOLUTION

7.**2**
We can make the following table from "the total amount of money kept in the three pouches in the first column of the third row is Rs. 4."

If the minimum and maximum value are 1, then the sum of the three pouches in the middle will be Rs 3.

	Column 1	Column 2	Column 3
Row 1			
Row 2		3 (1,1,1)	
Row 3	4 (1,1,2)		

If we calculate the maximum and minimum value possible for each slot in column 1. For the slot, column 1 and row 1, the maximum value possible is 10{2,4,4} while the minimum value possible is 8{2,2,4}.

Similarly, for the slot, column 1 and row 2, the maximum value possible is 13{3,5,5} while the minimum value possible is 11{3,3,5}.

It is known that the average amount of money (in rupees) kept in the nine pouches in any column or in any row is an integer. Thus the sum of coins in a row or column must be a multiple of 9.

So, we can iterate that 10,13,4 ...{27} is the only sum possible for the slots of column 1.

We now know two elements of row 2, thus we can iterate from the maximum and the minimum value possible for the slot {cloumn 3, row 2} that 38 is the only value possible for the slot.

We can make the following table:

	Column 1	Column 2	Column 3
Row 1	10 (2,4,4)		
Row 2	13 (3,5,5)	3 (1,1,1)	38 (6,12,20)
Row 3	4 (1,1,2)		

Similarly, we can find the amount for Column 2.

For the slot, column 2 and row 1, the maximum value possible is 22{6,8,8} while the minimum value possible is 20{6,6,8}.

For the slot, column 2 and row 3, the maximum value possible is 5{1,2,3} while the minimum value possible is 4{1,1,2}.

Thus {20,3,4} is the only solution possible.

	Column 1	Column 2	Column 3
Row 1	10 (2,4,4)	20 (6,6,8)	
Row 2	13 (3,5,5)	3 (1,1,1)	38 (6,12,20)
Row 3	4 (1,1,2)	4 (1,1,2)	

We can similarly make the following table for the last column.

	Column 1	Column 2	Column 3
Row 1	10 (2,4,4)	20 (6,6,8)	6 (1,2,3)
Row 2	13 (3,5,5)	3 (1,1,1)	38 (6,12,20)
Row 3	4 (1,1,2)	4 (1,1,2)	10 (2,3,5)

Answer 2

VIDEO SOLUTION

8. **C**
Let '100x' be the number of smartphones sold in year 2016.

Total revenue generated by Azra = 40x*15000 = Rs. 600000x

Total revenue generated by Bysi = 25x*20000 = Rs. 500000x

Total revenue generated by Cxqi = 15x*30000 = Rs. 450000x

Total revenue generated by Dipq = 20x*25000 = Rs. 500000x

We can see that revenue generated by Azra is the highest among all four brands. Hence, option C is the correct answer.

VIDEO SOLUTION

9. **C**
If at the starting point, the car was heading towards south then following would have occured : 1st 10 km in south ,2nd 10 km east, 3rd 20 km in south ,4th 40 km west, 5tht 10 km in south so overall 30 km to the west and 40 km to the south from starting point.

VIEW SOLUTION

10. **E**
Similar is the case with this question.Nothing can be inferred as we do not know about the fluctuation of share prices.

VIEW SOLUTION

11. **A**
Abdul gets the highest return since he buys all the shares at the start of the day.

Let Bikram and Chetan spend equal amounts of money. Since Chetan spends his money equally on all business, he would buy more of a shares at lower price and less shares at a higher price. On the other hand, Bikram buys an equal number of shares at each price. So, the return obtained by Chetan would be more than the return obtained by Bikram. Option a) is the correct answer.

VIEW SOLUTION

12. **A**

Now if after starting point at first point the 1st signal were 1 red and 2 green lights then distance traveled by the car would be as follows : 10 km in north then 10 km east then 20 km south then 40 km to west then 10 km to south . So overall the position of car is 30 km to the west and 20 km to south as compared to original position.

VIEW SOLUTION

13. **C**

According to given conditions, the car first travels 10 km north then 10 km to west then gain 20 km north then further travels east for 40 km then travel north for another 10 km before finally stopping. So in all the car travels 30 km to the east and 40 km north. Hence, calculating the distance radially we get 50 km in North-East direction.

VIEW SOLUTION

14. **A**

We want a diet with 10% minerals and at least 30% protien. The mixture of O and Q is suitable because both contain 10% minerals.In required proportion we can get 10% minerals and at least 30% proteins.
Solution P and R contains minerals which are less than 10% and maximum mineral %age in any component is 10. So if we include P or R in any mixture,we can't get 10% mineral %age.

So only 1 solution of O and Q is possible.

VIDEO SOLUTION

15. **8**

We can make the following table from "the total amount of money kept in the three pouches in the first column of the third row is Rs. 4."

If the minimum and maximum value are 1, then the sum of the three pouches in the middle will be Rs 3.

	Column 1	Column 2	Column 3
Row 1			
Row 2		3 (1,1,1)	
Row 3	4 (1,1,2)		

If we calculate the maximum and minimum value possible for each slot in column 1. For the slot, column 1 and row 1, the maximum value possible is 10{2,4,4} while the minimum value possible is 8{2,2,4}.

Similarly, for the slot, column 1 and row 2, the maximum value possible is 13{3,5,5} while the minimum value possible is 11{3,3,5}.

It is known that the average amount of money (in rupees) kept in the nine pouches in any column or in any row is an integer. Thus the sum of coins in a row or column must be a multiple of 9.

So, we can iterate that 10,13,4 ...{27} is the only sum possible for the slots of column 1.

We now know two elements of row 2, thus we can iterate from the maximum and the minimum value possible for the slot {cloumn 3, row 2} that 38 is the only value possible for the slot.

We can make the following table:

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For the slot, column 2 and row 1, the maximum value possible is 22{6,8,8} while the minimum value possible is 20{6,6,8}.

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	Column 1	Column 2	Column 3
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Row 3	4 (1,1,2)	4 (1,1,2)	

We can similarly make the following table for the last column.

	Column 1	Column 2	Column 3
Row 1	10 (2,4,4)	20 (6,6,8)	6 (1,2,3)
Row 2	13 (3,5,5)	3 (1,1,1)	38 (6,12,20)
Row 3	4 (1,1,2)	4 (1,1,2)	10 (2,3,5)

Answer 8

VIDEO SOLUTION

Cracku CAT Success Stories

16. **1**

		B	H	A	A	G	F
+		A	H	J	F	K	F
	A	A	F	G	C	A	F

The value of F can only be 0 as F+F=F can only hold if F=0.

Also, A can only be 1(in the second column) because to get a **carry** of more than 1, B has to be a double-digit number which is not possible. (**A carry is a digit that is transferred from one column of digits to another column of more significant digits**.)

So the data can be tabulated as follows:

		B	H	1	1	G	0
+		1	H	J	0	K	0
	1	1	0	G	C	1	0

Since the last row in the third column is 0, the carry to the second column must have been 1, Hence B+1+1=11 => B=9

In the 4th column, H+H = 10 since a carry 1 has gone to the 3rd column. Hence H=5.

G+K must be 11 and the carry 1 goes to the next column, so C=1+1=2.

Now, G,K can take values (3,8), (4,7) and (5,6) in any order.

From 5th column G=J+1 => J=G-1

Case: G=3 and K=8, here J =2 which is not possible as C =2

Case: G=8 and K=3, J=7, a possible case.

Case: G=4 and K=7, J=3 possible

Case: G=7 and K=4, J=6 possible

Case: G=5 and K=6, J=4 not possible as H =5.

Case: G=6 and K=5, J=5 both J and K are same, not possible.

Hence the cases can be tabulated as follows:

		9	5	1	1	8	0
+		1	5	7	0	3	0
	1	1	0	8	2	1	0

		9	5	1	1	7	0
+		1	5	6	0	4	0
	1	1	0	7	2	1	0

		9	5	1	1	4	0
+		1	5	3	0	7	0
	1	1	0	4	2	1	0

The letter A represents 1.

VIDEO SOLUTION

17. **9**

		B	H	A	A	G	F
+		A	H	J	F	K	F
	A	A	F	G	C	A	F

The value of F can only be 0 as F+F=F can only hold if F=0.

Also, A can only be 1(in the second column) because to get a **carry** of more than 1, B has to be a double-digit number which is not possible. (**A carry is a digit that is transferred from one column of digits to another column of more significant digits**.)

So the data can be tabulated as follows:

		B	H	1	1	G	0
+		1	H	J	0	K	0
	1	1	0	G	C	1	0

Since the last row in the third column is 0, the carry to the second column must have been 1, Hence B+1+1=11 => B=9

In the 4th column, H+H = 10 since a carry 1 has gone to the 3rd column. Hence H=5.

G+K must be 11 and the carry 1 goes to the next column, so C=1+1=2.

Now, G,K can take values (3,8), (4,7) and (5,6) in any order.

From 5th column G=J+1 => J=G-1

Case: G=3 and K=8, here J =2 which is not possible as C =2

Case: G=8 and K=3, J=7, a possible case.

Case: G=4 and K=7, J=3 possible

Case: G=7 and K=4, J=6 possible

Case: G=5 and K=6, J=4 not possible as H =5.

Case: G=6 and K=5, J=5 both J and K are same, not possible.

Hence the cases can be tabulated as follows:

		9	5	1	1	8	0
+		1	5	7	0	3	0
	1	1	0	8	2	1	0

		9	5	1	1	7	0
+		1	5	6	0	4	0
	1	1	0	7	2	1	0

		9	5	1	1	4	0
+		1	5	3	0	7	0
	1	1	0	4	2	1	0

The letter B represents 9.

VIDEO SOLUTION

18. **7**

		B	H	A	A	G	F
+		A	H	J	F	K	F
	A	A	F	G	C	A	F

The value of F can only be 0 as F+F=F can only hold if F=0.

Also, A can only be 1(in the second column) because to get a **carry** of more than 1, B has to be a double-digit number which is not possible. (**A carry is a digit that is transferred from one column of digits to another column of more significant digits**.)

So the data can be tabulated as follows:

		B	H	1	1	G	0
+		1	H	J	0	K	0
	1	1	0	G	C	1	0

Since the last row in the third column is 0, the carry to the second column must have been 1, Hence B+1+1=11 => B=9

In the 4th column, H+H = 10 since a carry 1 has gone to the 3rd column. Hence H=5.

G+K must be 11 and the carry 1 goes to the next column, so C=1+1=2.

Now, G,K can take values (3,8), (4,7) and (5,6) in any order.

From 5th column G=J+1 => J=G-1

Case: G=3 and K=8, here J =2 which is not possible as C =2

Case: G=8 and K=3, J=7, a possible case.

Case: G=4 and K=7, J=3 possible

Case: G=7 and K=4, J=6 possible

Case: G=5 and K=6, J=4 not possible as H =5.

Case: G=6 and K=5, J=5 both J and K are same, not possible.

Hence the cases can be tabulated as follows:

		9	5	1	1	8	0
+		1	5	7	0	3	0
	1	1	0	8	2	1	0

		9	5	1	1	7	0
+		1	5	6	0	4	0
	1	1	0	7	2	1	0

		9	5	1	1	4	0
+		1	5	3	0	7	0
	1	1	0	4	2	1	0

In all possible cases 7 is already represented by a letter other than D. Hence 7 is the answer.

VIDEO SOLUTION

19. **6**

		B	H	A	A	G	F
+		A	H	J	F	K	F
	A	A	F	G	C	A	F

The value of F can only be 0 as F+F=F can only hold if F=0.

Also, A can only be 1(in the second column) because to get a **carry** of more than 1, B has to be a double-digit number which is not possible. (**A carry is a digit that is transferred from one column of digits to another column of more significant digits**.)

So the data can be tabulated as follows:

		B	H	1	1	G	0
+		1	H	J	0	K	0
	1	1	0	G	C	1	0

Since the last row in the third column is 0, the carry to the second column must have been 1, Hence B+1+1=11 => B=9

In the 4th column, H+H = 10 since a carry 1 has gone to the 3rd column. Hence H=5.

G+K must be 11 and the carry 1 goes to the next column, so C=1+1=2.

Now, G,K can take values (3,8), (4,7) and (5,6) in any order.

From 5th column G=J+1 => J=G-1

Case: G=3 and K=8, here J =2 which is not possible as C =2

Case: G=8 and K=3, J=7, a possible case.

Case: G=4 and K=7, J=3 possible

Case: G=7 and K=4, J=6 possible

Case: G=5 and K=6, J=4 not possible as H =5.

Case: G=6 and K=5, J=5 both J and K are same, not possible.

Hence the cases can be tabulated as follows:

		9	5	1	1	8	0
+		1	5	7	0	3	0
	1	1	0	8	2	1	0

		9	5	1	1	7	0
+		1	5	6	0	4	0
	1	1	0	7	2	1	0

		9	5	1	1	4	0
+		1	5	3	0	7	0
	1	1	0	4	2	1	0

From the table it is clear that 6 cannot be represented by G.

VIDEO SOLUTION

20. **240**

It is given that the base exchange rates of A, B and C with respect to L are in the ratio 100:120:1. Let us assume that base exchange rates are '100a', '120a' and 'a' in that order.

It is given that the buying exchange rates of each of A, B, and C with respect to L are 5% below the corresponding base exchange rates. Therefore, we can say that the buying exchange rates are 95a, 114a, 0.95a.

It is given that the selling exchange rates of each of A, B, and C with respect to L are 10% above their corresponding base exchange rates. Therefore, we can say that the selling exchange rates are 110a, 132a, 1.1a.

We know about the opening and closing units in stock for each currency. Let us draw the table accordingly.

				No. of units			
Currency	**Buying rate**	**Base rate**	**Selling rate**	**Opening stock**	**Buy**	**Sell**	**Closing stock**
A	**95a**	**100a**	**110a**	**2500**			**3300**
B	**114a**	**120a**	**132a**	**4800**			**4800**
C	**0.95a**	**a**	**1.1a**	**48000**			**51000**

Let 'p', 'q' and 'r' be the number of units of currency A, B and C bought by the outlet on that day.

Then, we can say that the outlet sold 'p - 800', 'q' and 'r-3000' units of currency A, B and C respectively.

				No. of units			
Currency	**Buying rate**	**Base rate**	**Selling rate**	**Opening stock**	**Buy**	**Sell**	**Closing stock**
A	**95a**	**100a**	**110a**	**2500**	**p**	**p - 800**	**3300**
B	**114a**	**120a**	**132a**	**4800**	**q**	**q**	**4800**
C	**0.95a**	**a**	**1.1a**	**48000**	**r**	**r - 3000**	**51000**

It is given that the amount of L used by the outlet to buy C equals the amount of L it received by selling C.

$\Rightarrow$ 0.95a*r = 1.1a*(r - 3000)

$\Rightarrow$ 0.15r = 3300

$\Rightarrow$ r = 22000

It is also given that the amounts of L used by the outlet to buy A and B are in the ratio 5:3.

$$\Rightarrow \frac{p * 95a}{q * 114a} = \frac{5}{3}$$

$\Rightarrow$ p = 2q

Also, the amounts of L the outlet received from the sales of A and B are in the ratio 5:9.

$$\Rightarrow \frac{(p - 800) * 110a}{q * 132a} = \frac{5}{9}$$

$$\Rightarrow \frac{(2q - 800) * 110a}{q * 132a} = \frac{5}{9}$$

$\Rightarrow$ q = 600

Therefore, p = 2q = 2*600 = 1200.

It is given that the outlet received 88000 units of L by selling A during the day.

$\Rightarrow$ (p-800)*110a = 88000

$\Rightarrow$ (1200-800)*110a = 88000

$\Rightarrow$ 44000a = 88000

$\Rightarrow$ a = 2

We can fill the entire table and answer all the questions.

				No. of units			
Currency	Buying rate	Base rate	Selling rate	Opening stock	Buy	Sell	Closing stock
A	190	200	220	2500	1200	400	3300
B	228	240	264	4800	600	600	4800
C	1.9	2	2.2	48000	22000	19000	51000

From the table we can see that the base exchange rate of currency B with respect to currency L was 240.

VIDEO SOLUTION

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21. **A**

Let '100x' be the number of smartphones sold in year 2016. Then the number of smartphones sold in 2017 = 1.4*100x = 140x

It is given that Cxqi offered a 40% discount on its unit selling price in 2017 i.e. selling price in 2017 = 0.6*30000 = Rs. 18000

Also Cxqi's merket share increased by 15% whereas the other three brands lost 5% market share.

Brand	Maket Share (%)	No of units	Unit Selling Price (Rs.)	Profitibility(%)
Azra	35	49x	15,000	10
Bysi	20	28x	20,000	30
Cxqi	30	42x	18,000	20
Dipq	15	21x	25,000	30

Amount of profit generated by Azra = $\frac{10}{100}$*15000*49x = 73500x

Amount of profit generated by Bysi = $\frac{30}{100}$*20000*28x = 168000x

Amount of profit generated by Cxqi = $\frac{20}{100}$*18000*42x = 151200x

Amount of profit generated by Dipq = $\frac{30}{100}$*25000*21x = 157500x

We can see that brand Bysi generated maximum profit in year 2017. Hence, option A is the correct answer.

VIDEO SOLUTION

22. **A**

We are given the following table,

Country	GDP	GDP per capita	GDP growth rate	Population growth rate
Country 1	0.15	0.41	0.2%	-0.12%
Country 2	0.14	0.25	0.9%	-0.41%
Country 3	0.13	0.02	6.5%	0.70%
Country 4	0.12	0.38	0.5%	0.49%
Country 5	0.1	0.36	0.7%	0.31%
Country 6	0.08	0.08	3.2%	0.61%
Country 7	0.08	0.3	0.7%	-0.11%
Country 8	0.07	0.41	1.2%	0.71%

This table compares the GDP, GDP per capita, and population of eight countries, all of which are referenced using a consistent set of units.
Specifically, the GDP values are all given with the same country as the reference point (The reference point being Country 9)
Similarly, the GDP per capita among the eight countries in the table is given using the same reference point (The reference point being Country 10)
As a result, when comparing two countries from the list in terms of GDP, GDP per capita, or population, we can directly use the values presented in the table, since they are all based on a uniform reference point.

We can compare the population of two countries using the formula, $Population = \dfrac{GDP}{GDP\ Per\ Capita}$

For example, to compare the GDP of Country A and Country B, we can simply use the values listed under "GDP" in the table, as both are referenced using the same country throughout. This eliminates the need for additional conversions or adjustments.

We are given the GDP growth rate and we are told it is constant over the next three years,
Country 4 GDP: 0.12 and the given growth rate is 0.5%
GDP of Country 4 in 2026 will be $0.12 \times \ (1.005)^2$ this equals 0.121203

Country 4 GDP: 0.1 and the given growth rate is 3.2
GDP of Country 5 in 2026 will be $0.1 \times \ (1.007)^2$ this equals 0.1014049

Ratio will be, $\dfrac{0.121203}{0.1065024}$ this equals 1.19523

Hence, the answer is Option A

VIDEO SOLUTION

23.**2**
Neeta received least amount in bank deposits implies she received highest amount in property and the vice-versa for Geeta. The assets are 3 flats worth 90 lakh,a house worth 50 lakh, and a deposit worth 70 lakh. The total value of assets is 210 lakhs. They are divided equally, so each will receive assets worth 70 lakh.

No one daughter can get 3 flats as the total value of asset will be 90 lakhs which is greater than actual share.

All three daughters can't get 1-1 flat each as well. In that case, the daughter who owns 1 flat and the house will have assets worth 30+50 = 80 lakhs which is more than the actual share. Hence, we can conclude the one of the three daughter gets 2 flats and 10 lakhs bank deposit.

Out of the remaining two daughters, one will get the house and bank deposit worth 20 lakhs and the other one must have 1 flat and 40 lakhs in bank deposit. On the basis of bank distribution we can easily determine that property and bank deposits for each Neera, Seeta and Geeta.

	Neeta	Seeta	Geeta
Property	**2 flats worth 60 lakhs**	**house worth 50 lakhs**	**1 flat worth 30 lakhs**
Bank deposits	**10 lakhs**	**20 lakhs**	**40 lakhs**

From the table, we can see that Neeta received 2 flats.

VIDEO SOLUTION

24. **C**

Neeta received least amount in bank deposits implies she received highest amount in property and the vice-versa for Geeta. The assets are 3 flats worth 90 lakh,a house worth 50 lakh, and a deposit worth 70 lakh. The total value of assets is 210 lakhs. They are divided equally, so each will receive assets worth 70 lakh.

No one daughter can get 3 flats as the total value of asset will be 90 lakhs which is greater than actual share.

All three daughters can't get 1-1 flat each as well. In that case, the daughter who owns 1 flat and the house will have assets worth 30+50 = 80 lakhs which is more than the actual share. Hence, we can conclude the one of the three daughter gets 2 flats and 10 lakhs bank deposit.

Out of the remaining two daughters, one will get the house and bank deposit worth 20 lakhs and the other one must have 1 flat and 40 lakhs in bank deposit. On the basis of bank distribution we can easily determine that property and bank deposits for each Neera, Seeta and Geeta.

	Neeta	Seeta	Geeta
Property	**2 flats worth 60 lakhs**	**house worth 50 lakhs**	**1 flat worth 30 lakhs**
Bank deposits	**10 lakhs**	**20 lakhs**	**40 lakhs**

From the table, we can see that Seeta must have received Rs. 20 lakh in bank deposits. Hence, option C is the correct answer.

VIDEO SOLUTION

25. **13**

We can make the following table from "the total amount of money kept in the three pouches in the first column of the third row is Rs. 4."

If the minimum and maximum value are 1, then the sum of the three pouches in the middle will be Rs 3.

	Column 1	Column 2	Column 3
Row 1			
Row 2		3 (1,1,1)	
Row 3	4 (1,1,2)		

If we calculate the maximum and minimum value possible for each slot in column 1. For the slot, column 1 and row 1, the maximum value possible is 10{2,4,4} while the minimum value possible is 8{2,2,4}.

Similarly, for the slot, column 1 and row 2, the maximum value possible is 13{3,5,5} while the minimum value possible is 11{3,3,5}.

It is known that the average amount of money (in rupees) kept in the nine pouches in any column or in any row is an integer. Thus the sum of coins in a row or column must be a multiple of 9.

So, we can iterate that 10,13,4 ...{27} is the only sum possible for the slots of column 1.

We now know two elements of row 2, thus we can iterate from the maximum and the minimum value possible for the slot {cloumn 3, row 2} that 38 is the only value possible for the slot.

We can make the following table:

	Column 1	Column 2	Column 3
Row 1	10 (2,4,4)		
Row 2	13 (3,5,5)	3 (1,1,1)	38 (6,12,20)
Row 3	4 (1,1,2)		

Similarly, we can find the amount for Column 2.

For the slot, column 2 and row 1, the maximum value possible is 22{6,8,8} while the minimum value possible is 20{6,6,8}.

For the slot, column 2 and row 3, the maximum value possible is 5{1,2,3} while the minimum value possible is 4{1,1,2}.

Thus {20,3,4} is the only solution possible.

	Column 1	Column 2	Column 3
Row 1	10 (2,4,4)	20 (6,6,8)	
Row 2	13 (3,5,5)	3 (1,1,1)	38 (6,12,20)
Row 3	4 (1,1,2)	4 (1,1,2)	

We can similarly make the following table for the last column.

	Column 1	Column 2	Column 3
Row 1	10 (2,4,4)	20 (6,6,8)	6 (1,2,3)
Row 2	13 (3,5,5)	3 (1,1,1)	38 (6,12,20)
Row 3	4 (1,1,2)	4 (1,1,2)	10 (2,3,5)

The total amount of money (in rupees) in the three pouches kept in the first column of the second row=13

Correct answer 13

26. **A**

We are given the following table,

Country	GDP	GDP per capita	GDP growth rate	Population growth rate
Country 1	0.15	0.41	0.2%	-0.12%
Country 2	0.14	0.25	0.9%	-0.41%
Country 3	0.13	0.02	6.5%	0.70%
Country 4	0.12	0.38	0.5%	0.49%
Country 5	0.1	0.36	0.7%	0.31%
Country 6	0.08	0.08	3.2%	0.61%
Country 7	0.08	0.3	0.7%	-0.11%
Country 8	0.07	0.41	1.2%	0.71%

This table compares the GDP, GDP per capita, and population of eight countries, all of which are referenced using a consistent set of units.
Specifically, the GDP values are all given with the same country as the reference point (The reference point being Country 9)
Similarly, the GDP per capita among the eight countries in the table is given using the same reference point (The reference point being Country 10)
As a result, when comparing two countries from the list in terms of GDP, GDP per capita, or population, we can directly use the values presented in the table, since they are all based on a uniform reference point.

We can compare the population of two countries using the formula, $Population = \frac{GDP}{GDP\ Per\ Capita}$

For example, to compare the GDP of Country A and Country B, we can simply use the values listed under "GDP" in the table, as both are referenced using the same country throughout. This eliminates the need for additional conversions or adjustments.

In this question we are asked to find which of the countries among the four options has the least population in 2024 (The current year)
Option A: Country 8
GDP=0.07 and GDP Per Capita=0.41
Population=0.07/0.41 or $\frac{7}{41}$

Option B: Country 3
GDP=0.13 and GDP Per Capita=0.02
Population=0.13/0.02 or $\frac{13}{2}$

Option C: Country 7
GDP=0.08 and GDP Per Capita=0.3
Population=0.08/0.3 or $\frac{8}{30}$

Option D: Country 5
GDP=0.1 and GDP Per Capita=0.36
Population=0.1/0.36 or $\frac{10}{36}$

Comparing the four values, we are looking for the smallest values among the four,
$\frac{7}{41}, \frac{13}{2}, \frac{8}{30}, \frac{10}{36}$

$\frac{7}{41}$ is the smallest value among the four, hence Option A or Country 8 is the answer.

VIDEO SOLUTION

27. **0**
We are given the following table,

Country	GDP	GDP per capita	GDP growth rate	Population growth rate
Country 1	0.15	0.41	0.2%	-0.12%
Country 2	0.14	0.25	0.9%	-0.41%
Country 3	0.13	0.02	6.5%	0.70%
Country 4	0.12	0.38	0.5%	0.49%
Country 5	0.1	0.36	0.7%	0.31%
Country 6	0.08	0.08	3.2%	0.61%
Country 7	0.08	0.3	0.7%	-0.11%
Country 8	0.07	0.41	1.2%	0.71%

This table compares the GDP, GDP per capita, and population of eight countries, all of which are referenced using a consistent set of units.
Specifically, the GDP values are all given with the same country as the reference point (The reference point being Country 9)
Similarly, the GDP per capita among the eight countries in the table is given using the same reference point (The reference point being Country 10)
As a result, when comparing two countries from the list in terms of GDP, GDP per capita, or population, we can directly use the values presented in the table, since they are all based on a uniform reference point.

We can compare the population of two countries using the formula, $Population = \frac{GDP}{GDP\ Per\ Capita}$

For example, to compare the GDP of Country A and Country B, we can simply use the values listed under "GDP" in the table, as both are referenced using the same country throughout. This eliminates the need for additional conversions or adjustments.

We are asked to find the number of countries where the GDP per capita is lower in 2027 than it was in 2024.

For the GDP per capita to be lower in the consequent years, the population growth rate has to exceed the GDP growth rate, since GDP per capita is nothing but GDP divided by the population. That means if the population growth rate is lesser than that of the GDP growth rate, the GDP per capita will only increase.

We can clearly see that none of the following countries fall in the scenario where GDP growth rate is lesser than that of the population growth rate. Countries 1, 2 and 7 have actually decreasing population rates thereby definitely increasing the GDP per capita.
The rest of the countries have GDP growth rates larger than the population growth rates.

Hence, we can conclude that, none of the countries will have a smaller GDP per capita in 2027 when compared to 2024.

VIDEO SOLUTION

28. **B**
It is given that the base exchange rates of A, B and C with respect to L are in the ratio 100:120:1. Let us assume that base exchange rates are '100a', '120a' and 'a' in that order.

It is given that the buying exchange rates of each of A, B, and C with respect to L are 5% below the corresponding base exchange rates. Therefore, we can say that the buying exchange rates are 95a, 114a, 0.95a.

It is given that the selling exchange rates of each of A, B, and C with respect to L are 10% above their corresponding base exchange rates. Therefore, we can say that the selling exchange rates are 110a, 132a, 1.1a.

We know about the opening and closing units in stock for each currency. Let us draw the table accordingly.

				No. of units			
Currency	**Buying rate**	**Base rate**	**Selling rate**	**Opening stock**	**Buy**	**Sell**	**Closing stock**
A	**95a**	**100a**	**110a**	**2500**			**3300**
B	**114a**	**120a**	**132a**	**4800**			**4800**
C	**0.95a**	**a**	**1.1a**	**48000**			**51000**

Let 'p', 'q' and 'r' be the number of units of currency A, B and C bought by the outlet on that day.

Then, we can say that the outlet sold 'p - 800', 'q' and 'r-3000' units of currency A, B and C respectively.

				No. of units			
Currency	Buying rate	Base rate	Selling rate	Opening stock	Buy	Sell	Closing stock
A	95a	100a	110a	2500	p	p - 800	3300
B	114a	120a	132a	4800	q	q	4800
C	0.95a	a	1.1a	48000	r	r - 3000	51000

It is given that the amount of L used by the outlet to buy C equals the amount of L it received by selling C.

$\Rightarrow$ 0.95a*r = 1.1a*(r - 3000)

$\Rightarrow$ 0.15r = 3300

$\Rightarrow$ r = 22000

It is also given that the amounts of L used by the outlet to buy A and B are in the ratio 5:3.

$$\Rightarrow \frac{p*95a}{q*114a} = \frac{5}{3}$$

$\Rightarrow$ p = 2q

Also, the amounts of L the outlet received from the sales of A and B are in the ratio 5:9.

$$\Rightarrow \frac{(p-800)*110a}{q*132a} = \frac{5}{9}$$

$$\Rightarrow \frac{(2q-800)*110a}{q*132a} = \frac{5}{9}$$

$\Rightarrow$ q = 600

Therefore, p = 2q = 2*600 = 1200.

It is given that the outlet received 88000 units of L by selling A during the day.

$\Rightarrow$ (p-800)*110a = 88000

$\Rightarrow$ (1200-800)*110a = 88000

$\Rightarrow$ 44000a = 88000

$\Rightarrow$ a = 2

We can fill the entire table and answer all the questions.

				No. of units			
Currency	Buying rate	Base rate	Selling rate	Opening stock	Buy	Sell	Closing stock
A	190	200	220	2500	1200	400	3300
B	228	240	264	4800	600	600	4800
C	1.9	2	2.2	48000	22000	19000	51000

From the table we can see that the currency outlet sold 19000 units of currency C. Hence, option B is the correct answer.

VIDEO SOLUTION

29. **A**

Let us make note of the information given in the set.
The set states that rural kids were surveyed from 3 regions NE, West and South.
50 students each were surveyed from 150 villages in the NE, 250 villages from the West and 200 villages from the South.

Total number of kids from the NE = 150*50 = 7500
Total number of kids from the West = 250*50 = 12500
Total number of kids from the South = 200*50 = 10,000.

The table, given in the question, gives the number of students whose mothers dropped out before completing primary education. Therefore, the mothers of the remaining students should have completed primary education.

There are 7500 students from the NE in total. The mothers of 5000 of those students dropped out before completing primary education. Therefore, the remaining 2500 kids should have mothers who have completed primary education.

Let us make 2 tables - one representing the number of students and the other representing the number of students whose mother has completed primary education.

55% of the surveyed kids studied in schools run by the Government.
Number of kids studying in Govt. School should be 0.55*30000 = 16,500

From the given table, we know that the mothers of 13,500 kids who went to Govt. Schools dropped out of primary school.
Therefore, the remaining 16,500 - 13,500 = 3000 kids should have mother who have completed primary school.

37% of the surveyed kids study in private schools.
Number of kids studying in private schools should be 0.37*30000 = 11,100.
Number of kids in private schools whose mothers have completed primary school = 11,100 - 2700 = 8400.

8% of the surveyed kids did not go to school.
Number of kids who did not go to school = 0.08* 30000 = 2400
Number of kids not going to schools whose mothers have completed primary school = 2400 - 1800 = 600.

In S, 60% of the surveyed kids were in G.
Therefore, 6000 kids in S must be from G.

In NE, among the O kids, 50% had mothers who had dropped out before completing primary education.
There are 300 O kids whose mothers dropped out before completing primary education. These kids represent 50% of the total number of O-kids. Therefore, there must be 600 O-kids from NE - 300 kids should have mothers who dropped out before completing primary education and 300 kids should have mothers who have completed primary education.

The number of kids in G in NE was the same as the number of kids in G in W. Therefore, the number of kids in G in NE and the number of kids in G in W should be equal to 5250. Total number of kids:

	G	P	O	Total
NE	5250	1650	600	7500
W	5250			12500
S	6000			10000
Total	16,500	11,100	2,400	30,000

Number of kids whose mothers have completed primary education:

	G	P	O	Total
NE	1050	1150	300	2500
W	1050			5200
S	900			4300
Total	3000	8400	600	12000

We have been given that in S, all surveyed kids whose mothers had completed primary education were in school. Therefore, the number of kids not going to school whose mothers have completed primary education in S should be 0. Filling the tables, we get,

Total number of kids:

	G	P	O	Total
NE	5250	1650	600	7500
W	5250	5750	1500	12500
S	6000	3700	300	10000
Total	16,500	11,100	2,400	30,000

Number of kids whose mothers have completed primary education:

	G	P	O	Total
NE	1050	1150	300	2500
W	1050	3850	300	5200
S	900	3400	0	4300
Total	3000	8400	600	12000

As we can see from the table, 3700 kids out of the 10,000 kids from S are studying in P. 37% of the total number of students from S were studying in P. Therefore, option A is the right answer.

VIDEO SOLUTION

30. **6**

If a customer gives 500 rupee notes as her preferred denomination, the number of 500 rupee notes dispensed must be greater than the number of the notes of other denominations dispensed.

If Rs.3500 is dispensed as 500 rupee notes (7 notes), the remaining 1500 rupees should be dispensed using Rs.100 and Rs.200 notes. The minimum number of notes of other denomination required in this case will be 8 (7*200 + 1*100). Therefore, at least Rs.4000 should be dispensed as 500 rupee notes.

Case (1):
Rs.4000 is dispensed using 500 rupee notes, 8 five hundred rupee notes will be dispensed.
The remaining 1000 rupees cannot be fully dispensed as 100 rupee notes (since 10 notes will be required).
If 800 rupees is dispensed as 100 rupee notes, then 9 notes will be required to dispense 1000 rupees (8*100+200).
Therefore, we can eliminate these 2 cases.
If 600 rupees is dispensed using 100 rupee notes, then a minimum of 8 notes will be required to dispense 1000 rupees (6*100 + 2*200). Therefore, we can eliminate this case as well.

If 400 rupees is dispensed using 100 rupee notes, then 7 notes will be required (4*100+3*200). This is a valid case.
If 200 rupees is dispensed using 100 rupee notes, then 6 notes will be required (2*100+4*200). This is a valid case.
1000 rupees can be dispensed using 5 notes of Rs.200.

Therefore, there are 3 valid cases.

Case (2):
Rs.4500 is dispensed using 500 rupee notes. 9 five hundred rupee notes will be dispensed in this case.
The remaining 500 rupees can be dispensed as 100 rupee notes (5 notes) or a combination of 100 rupee and 200 rupee notes.
200*a + 100*b = 500
'a' can take 0, 1, and 2.
Therefore, there are 3 valid cases.

Case (3):
5000 rupees is dispensed using 10 five hundred rupee notes.
There is only 1 valid case.

It has been given that the ATM could serve only 10 customers with a stock of fifty 500 rupee notes. We have to find the maximum number of customers who could have given Rs.500 as their preference.

The least number of 500 rupee notes required to serve a customer who has given Rs.500 as the preference is 8. Using 50 five hundred rupee notes, we can serve [50/8] = 6 customers. Therefore, 6 is the correct answer.

VIDEO SOLUTION

31. **A**

It has been given that the total number of notes dispensed is the smallest possible. Therefore, we have to minimize the number of notes dispensed in each of the 2 cases given.

The least number of notes required to serve a customer who has given 500 rupees as his preference is 10.
50 customers who have given 500 rupee notes as their preference have to be served. We will require 50*10 = 500 notes for this purpose.

Let us consider the case when a customer has given Rs.100 as his preference.
As we have seen, minimum number of notes will be required when we maximize the number of five hundred rupee notes as much as possible.

If Rs.4000 is dispensed using 500 rupee notes, the remaining 1000 rupees can be dispensed using ten 100 rupee notes. In this case, the number of 100 rupee notes (10) is greater than the number of 500 rupee notes (8). This is a valid case. We have to find if we can reduce the number of notes required any further.

We cannot increase the number of 500 rupee notes to 9 since only 5 hundred rupee notes can be dispensed, violating the condition that the customer has given 100 as his preferred denomination.

If we replace two 100 rupee notes with one 200 rupee note, then the number of 100 rupee notes will become 6. The number of 500 rupee notes (8) exceeds the number of 100 rupee note (6). Therefore, dispensing 4000 rupees using 500 rupee notes and the rest using 100 rupee notes represents the optimum condition.

The minimum number of notes required to serve 1 customer = 8 (five hundred notes) + 10 (hundred notes) = 18
Number of five hundred notes required to serve 50 customers = 8*50 = 400

Therefore, the total number of notes required = 400 + 500 = 900.
Therefore, option A is the right answer.

VIDEO SOLUTION

32. **3**

We can make the following table from "the total amount of money kept in the three pouches in the first column of the third row is Rs. 4."

If the minimum and maximum value are 1, then the sum of the three pouches in the middle will be Rs 3.

	Column 1	Column 2	Column 3
Row 1			
Row 2		3 (1,1,1)	
Row 3	4 (1,1,2)		

If we calculate the maximum and minimum value possible for each slot in column 1. For the slot, column 1 and row 1, the maximum value possible is 10{2,4,4} while the minimum value possible is 8{2,2,4}.

Similarly, for the slot, column 1 and row 2, the maximum value possible is 13{3,5,5} while the minimum value possible is 11{3,3,5}.

It is known that the average amount of money (in rupees) kept in the nine pouches in any column or in any row is an integer. Thus the sum of coins in a row or column must be a multiple of 9.

So, we can iterate that 10,13,4 ...{27} is the only sum possible for the slots of column 1.

We now know two elements of row 2, thus we can iterate from the maximum and the minimum value possible for the slot {cloumn 3, row 2} that 38 is the only value possible for the slot.

We can make the following table:

	Column 1	Column 2	Column 3
Row 1	10 (2,4,4)		
Row 2	13 (3,5,5)	3 (1,1,1)	38 (6,12,20)
Row 3	4 (1,1,2)		

Similarly, we can find the amount for Column 2.

For the slot, column 2 and row 1, the maximum value possible is 22{6,8,8} while the minimum value possible is 20{6,6,8}.

For the slot, column 2 and row 3, the maximum value possible is 5{1,2,3} while the minimum value possible is 4{1,1,2}.

Thus {20,3,4} is the only solution possible.

	Column 1	Column 2	Column 3
Row 1	10 (2,4,4)	20 (6,6,8)	
Row 2	13 (3,5,5)	3 (1,1,1)	38 (6,12,20)
Row 3	4 (1,1,2)	4 (1,1,2)	

We can similarly make the following table for the last column.

	Column 1	Column 2	Column 3
Row 1	10 (2,4,4)	20 (6,6,8)	6 (1,2,3)
Row 2	13 (3,5,5)	3 (1,1,1)	38 (6,12,20)
Row 3	4 (1,1,2)	4 (1,1,2)	10 (2,3,5)

Answer 3

VIDEO SOLUTION

33. **D**
We want the minimum vote share, that means both the candidates run staid campaigns.
And, both the campaigns should be attacking, since we see that if one candidate runs an attacking campaign whereas the other candidate runs an issues campaign, a fair number of voters get transferred to the other candidate's vote share.

This points to us to a scenario where both the campaigns are staid and attacking, which is nothing but the scenario described in the previous question.

If both Ramya and Amiya run staid campaigns, the intensity of each staid campaign is 1
So total number of voters that would vote for them if they focused on issues will be 20x(1+1)%=40%

This means if they had both ran regarding issues, then they would get 20% of votes each.
We are told that both of them run attacking campaigns,
And the rule for mutual attacking campaign is 10% of voters who would have voted for each candidate will not vote.
That means 10% of 20% of each candidate will not vote, now that it is a mutually attacking campaign.
That means, each candidate receives 18% of the votes.

Total votes received is 36%, which is the minimum possible.

VIDEO SOLUTION

34. **E**
Since we do not know the exact prices of the share,we cannot determine who gained the most,

VIEW SOLUTION

35. **A**
Let us make note of the information given in the set.
The set states that rural kids were surveyed from 3 regions NE, West and South.
50 students each were surveyed from 150 villages in the NE, 250 villages from the West and 200 villages from the South.

Total number of kids from the NE = 150*50 = 7500
Total number of kids from the West = 250*50 = 12500
Total number of kids from the South = 200*50 = 10,000.

The table, given in the question, gives the number of students whose mothers dropped out before completing primary education. Therefore, the mothers of the remaining students should have completed primary education.

There are 7500 students from the NE in total. The mothers of 5000 of those students dropped out before completing primary education. Therefore, the remaining 2500 kids should have mothers who have completed primary education.

Let us make 2 tables - one representing the number of students and the other representing the number of students whose mother has completed primary education.

55% of the surveyed kids studied in schools run by the Government.
Number of kids studying in Govt. School should be 0.55*30000 = 16,500

From the given table, we know that the mothers of 13,500 kids who went to Govt. Schools dropped out of primary school.
Therefore, the remaining 16,500 - 13,500 = 3000 kids should have mother who have completed primary school.

37% of the surveyed kids study in private schools.
Number of kids studying in private schools should be 0.37*30000 = 11,100.
Number of kids in private schools whose mothers have completed primary school = 11,100 - 2700 = 8400.

8% of the surveyed kids did not go to school.
Number of kids who did not go to school = 0.08* 30000 = 2400
Number of kids not going to schools whose mothers have completed primary school = 2400 - 1800 = 600.

In S, 60% of the surveyed kids were in G.
Therefore, 6000 kids in S must be from G.

In NE, among the O kids, 50% had mothers who had dropped out before completing primary education.
There are 300 O kids whose mothers dropped out before completing primary education. These kids represent 50% of the total number of O-kids. Therefore, there must be 600 O-kids from NE - 300 kids should have mothers who dropped out before completing primary education and 300 kids should have mothers who have completed primary education.

The number of kids in G in NE was the same as the number of kids in G in W. Therefore, the number of kids in G in NE and the number of kids in G in W should be equal to 5250. Total number of kids:

	G	P	O	Total
NE	5250	1650	600	7500
W	5250			12500
S	6000			10000
Total	16,500	11,100	2,400	30,000

Number of kids whose mothers have completed primary education:

	G	P	O	Total
NE	1050	1150	300	2500
W	1050			5200
S	900			4300
Total	3000	8400	600	12000

We have been given that in S, all surveyed kids whose mothers had completed primary education were in school. Therefore, the number of kids not going to school whose mothers have completed primary education in S should be 0. Filling the tables, we get,

Total number of kids:

	G	P	O	Total
NE	5250	1650	600	7500
W	5250	5750	1500	12500
S	6000	3700	300	10000
Total	16,500	11,100	2,400	30,000

Number of kids whose mothers have completed primary education:

	G	P	O	Total
NE	1050	1150	300	2500
W	1050	3850	300	5200
S	900	3400	0	4300
Total	3000	8400	600	12000

Number of kids who were out of school in the initial survey = 2400.
The number of kids who were out of school in NE is 600, W is 1500 and S is 300. 25% of the kids from one area, 100% of the kids from one area and 50% of the kids from another area are transferred to G. As a result, the number of kids who were not in school earlier but studying in G now became 50% of 2400 = 1200.

100% of the kids in W could not have been transferred to G (Since the total number of kids transferred to G is 1200 and the number of kids in W alone is 1500).

Let us assume that 100% of the kids from NE transferred (600). In this case, if 25% of the kids from W are transferred (375), 50% of the kids from S (300) will be transferred. The total in this case would be 600 + 375+ 300 = 1275. Therefore, we can eliminate this case.

If 100% of the kids from NE (600), 25% of the kids from S (75) and 50% of the kids from W (750) are transferred, then the total would have been 1425. Therefore, we can eliminate this case as well.

If 50% of the kids from W (750), 25% of the kids from NE (150) and 100% of the kids from S (300) are transferred, then the total number of kids transferred to G would have been 750 + 150 + 300 = 1200. Therefore, this must have been the case.

Number of students transferred to G in W = 750.

Number of students present in G in W earlier = 5250

Total = 6000

Therefore, option A is the right answer.

VIDEO SOLUTION

36. **D**
Since, we have been given that a person can be reached only by those who are smaller than him. Hence, 1 cannot be reached by anyone. Thus, option D is the correct answer.

VIDEO SOLUTION

37. **C**
Let us go by options.

Option A: In first row, 6 is reached by 3, 1 by none, 2 by none, 4 by 3, 3 by none. Hence, ruled out.

Option B: Third row, every individual is approached by other.

Option D: In first row, 6 is reached by 3, 1 by none, 2 by none, 4 by 3, 3 by none. Hence, ruled out.

Hence, Option C is correct.

VIDEO SOLUTION

38. **E**
Let the share prices at 10,11,12,1,2,3 be p,q,r,s,t,u
Since Abdul lost all money, p>u
From Dane it can be observed, s+t+u>p+q+r
p>r and u>t
From Emily it can be observed, r+u>p+s

Thus we get sequence p > u > t > q and r > s.

From these inequalities it can be inferred that the p is the highest

Since t > q; option A is definitely false.

Also r>s so option D is definitely false.

VIEW SOLUTION

39. **7**

It has been given that the customer gives 500 rupee notes as her preferred denomination.
Therefore, the number of 500 rupee notes dispensed must be greater than the number of the notes of other denominations dispensed.

If Rs.3500 is dispensed as 500 rupee notes (7 notes), the remaining 1500 rupees should be dispensed using Rs.100 and Rs.200 notes. The minimum number of notes of other denomination required in this case will be 8 (7*200 + 1*100). Therefore, at least Rs.4000 should be dispensed as 500 rupee notes.

Case (1):
Rs.4000 is dispensed using 500 rupee notes, 8 five hundred rupee notes will be dispensed.
The remaining 1000 rupees cannot be fully dispensed as 100 rupee notes (since 10 notes will be required).
If 800 rupees is dispensed as 100 rupee notes, then 9 notes will be required to dispense 1000 rupees (8*100+200).
Therefore, we can eliminate these 2 cases.
If 600 rupees is dispensed using 100 rupee notes, then a minimum of 8 notes will be required to dispense 1000 rupees (6*100 + 2*200). Therefore, we can eliminate this case as well.

If 400 rupees is dispensed using 100 rupee notes, then 7 notes will be required (4*100+3*200). This is a valid case.
If 200 rupees is dispensed using 100 rupee notes, then 6 notes will be required (2*100+4*200). This is a valid case.
1000 rupees can be dispensed using 5 notes of Rs.200.

Therefore, there are 3 valid cases.

Case (2):
Rs.4500 is dispensed using 500 rupee notes. 9 five hundred rupee notes will be dispensed in this case.
The remaining 500 rupees can be dispensed as 100 rupee notes (5 notes) or a combination of 100 rupee and 200 rupee notes.
200*a + 100*b = 500
'a' can take 0, 1, and 2.
Therefore, there are 3 valid cases.

Case (3):
5000 rupees is dispensed using 10 five hundred rupee notes.
There is only 1 valid case.

Total number of valid cases = 3+3+1 = 7.
Therefore, 7 is the right answer.

VIDEO SOLUTION

40. **E**

Two ingredients are mixed in equal proportion. So, the minimum amount of protein and carbohydrate should be 60 each, the minimum amount of minerals is 10 and the maximum amount of fat is 50. The combination of O and S satisfies this requirement.

VIDEO SOLUTION

CAT Previous Papers PDF

41. **A**
The distance traveled by the car in particular direction is as follows : First 10 km in north then 10 km in west then 20 km north then 40 km east then again 10 km north.

So in all total distance traveled is 10+10+20+40+10 = 90 kms.

VIEW SOLUTION

42. **A**
Let us make note of the information given in the set.
The set states that rural kids were surveyed from 3 regions NE, West and South.
50 students each were surveyed from 150 villages in the NE, 250 villages from the West and 200 villages from the South.

Total number of kids from the NE = 150*50 = 7500
Total number of kids from the West = 250*50 = 12500
Total number of kids from the South = 200*50 = 10,000.

The table, given in the question, gives the number of students whose mothers dropped out before completing primary education. Therefore, the mothers of the remaining students should have completed primary education.

There are 7500 students from the NE in total. The mothers of 5000 of those students dropped out before completing primary education. Therefore, the remaining 2500 kids should have mothers who have completed primary education.

Let us make 2 tables - one representing the number of students and the other representing the number of students whose mother has completed primary education.

55% of the surveyed kids studied in schools run by the Government.
Number of kids studying in Govt. School should be 0.55*30000 = 16,500

From the given table, we know that the mothers of 13,500 kids who went to Govt. Schools dropped out of primary school.
Therefore, the remaining 16,500 - 13,500 = 3000 kids should have mother who have completed primary school.

37% of the surveyed kids study in private schools.
Number of kids studying in private schools should be 0.37*30000 = 11,100.
Number of kids in private schools whose mothers have completed primary school = 11,100 - 2700 = 8400.

8% of the surveyed kids did not go to school.
Number of kids who did not go to school = 0.08* 30000 = 2400
Number of kids not going to schools whose mothers have completed primary school = 2400 - 1800 = 600.

In S, 60% of the surveyed kids were in G.
Therefore, 6000 kids in S must be from G.

In NE, among the O kids, 50% had mothers who had dropped out before completing primary education.
There are 300 O kids whose mothers dropped out before completing primary education. These kids represent

50% of the total number of O-kids. Therefore, there must be 600 O-kids from NE - 300 kids should have mothers who dropped out before completing primary education and 300 kids should have mothers who have completed primary education.

The number of kids in G in NE was the same as the number of kids in G in W. Therefore, the number of kids in G in NE and the number of kids in G in W should be equal to 5250. Total number of kids:

	G	P	O	Total
NE	5250	1650	600	7500
W	5250			12500
S	6000			10000
Total	16,500	11,100	2,400	30,000

Number of kids whose mothers have completed primary education:

	G	P	O	Total
NE	1050	1150	300	2500
W	1050			5200
S	900			4300
Total	3000	8400	600	12000

We have been given that in S, all surveyed kids whose mothers had completed primary education were in school. Therefore, the number of kids not going to school whose mothers have completed primary education in S should be 0. Filling the tables, we get,

Total number of kids:

	G	P	O	Total
NE	5250	1650	600	7500
W	5250	5750	1500	12500
S	6000	3700	300	10000
Total	16,500	11,100	2,400	30,000

Number of kids whose mothers have completed primary education:

	G	P	O	Total
NE	1050	1150	300	2500
W	1050	3850	300	5200
S	900	3400	0	4300
Total	3000	8400	600	12000

Number of kids who were out of school in the initial survey = 2400.
The number of kids who were out of school in NE is 600, W is 1500 and S is 300. 25% of the kids from one area, 100% of the kids from one area and 50% of the kids from another area are transferred to G. As a result, the number of kids who were not in school earlier but studying in G now became 50% of 2400 = 1200.

100% of the kids in W could not have been transferred to G (Since the total number of kids transferred to G is 1200 and the number of kids in W alone is 1500).

Let us assume that 100% of the kids from NE transferred (600). In this case, if 25% of the kids from W are transferred (375), 50% of the kids from S (300) will be transferred. The total in this case would be 600 + 375+ 300 = 1275. Therefore, we can eliminate this case.

If 100% of the kids from NE (600), 25% of the kids from S (75) and 50% of the kids from W (750) are transferred, then the total would have been 1425. Therefore, we can eliminate this case as well.

If 50% of the kids from W (750), 25% of the kids from NE (150) and 100% of the kids from S (300) are transferred, then the total number of kids transferred to G would have been 750 + 150 + 300 = 1200. Therefore, this must have been the case.

Already, 5100 kids whose mothers had dropped out were in G in S.
After the transfer, 300 more students will be added to the count.
Therefore, 5400 students whose mothers had dropped out will be in G in S.
Total number of kids whose mothers had dropped out in S = 5700

Percentage = 5400/5700 = 94.73%.

Therefore, option A is the right answer.

VIDEO SOLUTION

43. **20**

Let the total number of gold coins with the old woman be '9n'.
Total value of the assets with the old woman = 50 + 3*30 + 70 + 9n = 210+9n.
We know that the assets have been distributed in the ratio 1:2:3.
Therefore, Neeta must have received 35+1.5n (by value), Seeta must have received 70+3n and Geeta must have received 105+4.5n.
Further, it has been given that the gold coins distributed were in the ratio 2:3:4.
Therefore, the number of gold coins with Neeta must be '2n', Seeta must be '3n' and Geeta must be '4n'.
Seeta has 3n gold coins. Therefore, the total value of the assets with her must be 70. Seeta could not have inherited all the flats. Therefore, Seeta must have received the house (worth 50 lakh) and 20 lakh from bank deposits.

We know that Geeta did not receive Rs. 30 lakh from the bank deposits. Therefore, Neeta must have received Rs. 30 lakh.
The remaining 5 lakh must be contributed by the gold coins (Since there is no other asset worth 5 lakh).
=> 5 + 1.5n = 2n
=> 0.5n = 5
=> n = 10
The oldwoman must have had 10*9 = 90 gold coins.
Total assets = 210 + 90*1 = 300 lakh
Neeta has received 50 lakh in total, Seeta has received 100 lakh and Geeta has received 150 lakh.
Geeta must have received 90 lakh from 3 flats. Out of the remaining 60 lakh, 4*10 = 40 lakh has been contributed by the gold coins. Geeta must have received 150 - 90 - 40 = 20 lakh from bank deposits. Therefore, 20 is the right answer.

VIDEO SOLUTION

44. **D**

It is given that the base exchange rates of A, B and C with respect to L are in the ratio 100:120:1. Let us assume that base exchange rates are '100a', '120a' and 'a' in that order.

It is given that the buying exchange rates of each of A, B, and C with respect to L are 5% below the corresponding base exchange rates. Therefore, we can say that the buying exchange rates are 95a, 114a, 0.95a.

It is given that the selling exchange rates of each of A, B, and C with respect to L are 10% above their corresponding base exchange rates. Therefore, we can say that the selling exchange rates are 110a, 132a, 1.1a.

We know about the opening and closing units in stock for each currency. Let us draw the table accordingly.

				No. of units			
Currency	**Buying rate**	**Base rate**	**Selling rate**	**Opening stock**	**Buy**	**Sell**	**Closing stock**
A	**95a**	**100a**	**110a**	**2500**			**3300**
B	**114a**	**120a**	**132a**	**4800**			**4800**
C	**0.95a**	**a**	**1.1a**	**48000**			**51000**

Let 'p', 'q' and 'r' be the number of units of currency A, B and C bought by the outlet on that day.

Then, we can say that the outlet sold 'p - 800', 'q' and 'r-3000' units of currency A, B and C respectively.

				No. of units			
Currency	**Buying rate**	**Base rate**	**Selling rate**	**Opening stock**	**Buy**	**Sell**	**Closing stock**
A	**95a**	**100a**	**110a**	**2500**	**p**	**p - 800**	**3300**
B	**114a**	**120a**	**132a**	**4800**	**q**	**q**	**4800**
C	**0.95a**	**a**	**1.1a**	**48000**	**r**	**r - 3000**	**51000**

It is given that the amount of L used by the outlet to buy C equals the amount of L it received by selling C.

$\Rightarrow$ 0.95a*r = 1.1a*(r - 3000)

$\Rightarrow$ 0.15r = 3300

$\Rightarrow$ r = 22000

It is also given that the amounts of L used by the outlet to buy A and B are in the ratio 5:3.

$$\Rightarrow \frac{p*95a}{q*114a} = \frac{5}{3}$$

$\Rightarrow$ p = 2q

Also, the amounts of L the outlet received from the sales of A and B are in the ratio 5:9.

$$\Rightarrow \frac{(p-800)*110a}{q*132a} = \frac{5}{9}$$

$$\Rightarrow \frac{(2q-800)*110a}{q*132a} = \frac{5}{9}$$

$\Rightarrow$ q = 600

Therefore, p = 2q = 2*600 = 1200.

It is given that the outlet received 88000 units of L by selling A during the day.

$\Rightarrow$ (p-800)*110a = 88000

$\Rightarrow$ (1200-800)*110a = 88000

$\Rightarrow$ 44000a = 88000

$\Rightarrow$ a = 2

We can fill the entire table and answer all the questions.

				No. of units			
Currency	**Buying rate**	**Base rate**	**Selling rate**	**Opening stock**	**Buy**	**Sell**	**Closing stock**
A	190	200	220	2500	1200	400	3300
B	228	240	264	4800	600	600	4800
C	1.9	2	2.2	48000	22000	19000	51000

From the table we can see that the buying exchange rate of currency C with respect to currency L was 1.9.
Hence, we can say that option D is the correct answer.

VIDEO SOLUTION

45. **A**

It has been given that the customer has to receive 20 notes at the maximum. Also, we have restriction on the number of 500 rupee notes (fifty) but we do not have any restriction on the number of notes of other denominations. Therefore, in order to serve the maximum number of customers, we have to minimize the number of 500 rupee notes dispensed as much as possible.

If no 500 rupee note is dispensed, then a minimum of 25 notes will be required (25 200 rupee notes).
If one 500 rupee note is dispensed, then a minimum of one 100 rupee note and twenty two 200 rupee notes will be required. The total number of notes required = 1 + 1 + 22 = 24. Therefore, we can eliminate this case.

If two 500 rupee notes are dispensed, then a minimum of 20 two hundred rupee notes will be required. We can eliminate this case as well since the number of notes required is greater than 20.

If three 500 rupee notes are dispensed, then a minimum of 1 hundred rupee note and 17 two hundred rupee notes will be required. The number of notes required in this case is 3+1+17 = 21. Therefore, we can eliminate this case as well.

If four 500 rupee notes are dispensed, then a minimum of 15 two hundred rupee notes will be required. Total number of notes required in this case is 4+15 = 19 < 20. Therefore, this is a valid case.

The least number of 500 rupee notes with which we can serve a customer such that the total number of notes dispensed does not exceed 20 is 4. Therefore, a maximum of [50/4] = 12 customers can be served with 50 five hundred rupee notes and hence, option A is the right answer.

46. **B**
We are looking for the minimum possible number of votes that Ramya can get and maximise the number of votes that Amiya can get.

We can borrow the scenario from the previous question where Ramya runs an attacking campaign, and we minimised the number of votes she can get.

To minimise the number of votes, we can have Ramya run a staid campaign to minimise the votes, so minimum intensity, which will get her 20% of the votes if she ran with issues. Now that she is running with attacking, she will loose 20% of the votes to Amiya and 5% of the votes will not vote anymore.

That is a total 25% loss. Remaining votes she will get is 75% of the 20% which will leave her with 15% of the votes.

And to maximise the number of votes Amiya can get, we will have her run an vigorous issues campaign, which will give her 2x20% of the votes, that is 40% of the votes. And since Ramya has been running an attacking campaign, 20% of her votes are transferred to Amiya. 20% of the 20% of the votes which is 4% that were going to Ramya will now go to Amiya. That will bring up Amiya's tally up to 44% leaving Ramya's tally at 15%.

The difference in the votes will be 44-15=29%.

This is the maximum possible vote difference between the two candidates that is possible.

47. **E**
P,Q,S contain 80%,10% and 45% carbohydrates respectively.
Option A,C and D does not contain 60% carbohydrates.

Option E: Carbohydrate %=[(4*80)+(1*10)+(1*45)]/6= 62.5%
Cost per unit=[(4*50)+(200)+100]/6=83.3

Option B: Carbohydrate %=[(4*80)+(1*20)+(2*45)]/7= 61.4%
Cost per unit=[(4*50)+(200)+(2*100)]/7=600/7=85.7

48. **C**
The table represent the platforms and by how many platforms can it be reached. For example, in row 1 and column 1, platform with height 6 can be reached by 3 people.

6(3)	1(0)	2(1)	4(3)	3(0)
9(4)	5(2)	3(2)	2(0)	8(5)
7(1)	8(4)	4(1)	6(6)	5(1)
3(1)	9(4)	5(3)	1(0)	2(1)
1(0)	7(2)	6(2)	3(1)	9(6)

Hence, there are a total of 7 people.

VIDEO SOLUTION

49. **A**

We are given the following table,

Country	GDP	GDP per capita	GDP growth rate	Population growth rate
Country 1	0.15	0.41	0.2%	-0.12%
Country 2	0.14	0.25	0.9%	-0.41%
Country 3	0.13	0.02	6.5%	0.70%
Country 4	0.12	0.38	0.5%	0.49%
Country 5	0.1	0.36	0.7%	0.31%
Country 6	0.08	0.08	3.2%	0.61%
Country 7	0.08	0.3	0.7%	-0.11%
Country 8	0.07	0.41	1.2%	0.71%

This table compares the GDP, GDP per capita, and population of eight countries, all of which are referenced using a consistent set of units.

Specifically, the GDP values are all given with the same country as the reference point (The reference point being Country 9)

Similarly, the GDP per capita among the eight countries in the table is given using the same reference point (The reference point being Country 10)

As a result, when comparing two countries from the list in terms of GDP, GDP per capita, or population, we can directly use the values presented in the table, since they are all based on a uniform reference point.

We can compare the population of two countries using the formula, $Population = \frac{GDP}{GDP\ Per\ Capita}$

For example, to compare the GDP of Country A and Country B, we can simply use the values listed under "GDP" in the table, as both are referenced using the same country throughout. This eliminates the need for additional conversions or adjustments.

The given question asks us to compare the population of countries between 1, 4, 5, and 7 in the year 2027

Country 1: Population in 2024 will be 0.15/0.41 or 15/41 and the population is decreasing at the rate 0.12%

Country 4: Population in 2024 will be 0.12/0.38 or 12/38 and the population is increasing at the rate of 0.49%

Country 5: Population in 2024 will be 0.1/0.36 or 10/36 and the population is increasing at the rate 0.31%

Country 7: Population in 2024 will be 0.08/0.3 or 8/30 and the population is decreasing at the rate 0.11%

Simplifying and comparing the four fractions in order, $\frac{15}{41}, \frac{6}{19}, \frac{5}{18}, \frac{4}{15}$

The decimal values of the following are: 0.36586, 0.315789, 0.2777, 0.2666

Right away we can eliminate Country 7 since the population is not only the least, but it is decreasing, so there is no chance it will have the highest population in three years time.

For the other three countries,

Country 1: $(0.36585) \times\ (0.9988)^3$ this equals 0.36453

Country 4: $(0.315789) \times\ (1.0049)^3$ this equals 0.32045

Country 7: $(0.2777) \times\ (1.0031)^3$ this equals 0.2803612

Hence, Country 1 will have the largest population among these four countries.

VIDEO SOLUTION

50. **D**
Option A: P and Q in 4:1 ratio gives fat 10% but the protein is 22%

Option B: P and S do not contain fat. Hence we wont analyse the option.

Option C: P and R in 3:1 ratio gives fat 10% but the protein is 27.5%

Option D: Q and S in 1:4 gives fat and protein as 10% and 46 %.

Cost of the mixture= [(1*200)+(4*100)]/5=120

Option E: R and S in 1:3 gives 10% fat and 50%protein.
Cost of the mixture= [(1*500)+(3*100)]/4=200

VIDEO SOLUTION

51. **1200**
It is given that the base exchange rates of A, B and C with respect to L are in the ratio 100:120:1. Let us assume that base exchange rates are '100a', '120a' and 'a' in that order.

It is given that the buying exchange rates of each of A, B, and C with respect to L are 5% below the corresponding base exchange rates. Therefore, we can say that the buying exchange rates are 95a, 114a, 0.95a.

It is given that the selling exchange rates of each of A, B, and C with respect to L are 10% above their corresponding base exchange rates. Therefore, we can say that the selling exchange rates are 110a, 132a, 1.1a.

We know about the opening and closing units in stock for each currency. Let us draw the table accordingly.

				No. of units			
Currency	Buying rate	Base rate	Selling rate	Opening stock	Buy	Sell	Closing stock
A	95a	100a	110a	2500			3300
B	114a	120a	132a	4800			4800
C	0.95a	a	1.1a	48000			51000

Let 'p', 'q' and 'r' be the number of units of currency A, B and C bought by the outlet on that day.

Then, we can say that the outlet sold 'p - 800', 'q' and 'r-3000' units of currency A, B and C respectively.

				No. of units			
Currency	Buying rate	Base rate	Selling rate	Opening stock	Buy	Sell	Closing stock
A	95a	100a	110a	2500	p	p - 800	3300
B	114a	120a	132a	4800	q	q	4800
C	0.95a	a	1.1a	48000	r	r - 3000	51000

It is given that the amount of L used by the outlet to buy C equals the amount of L it received by selling C.

$\Rightarrow$ 0.95a*r = 1.1a*(r - 3000)

$\Rightarrow$ 0.15r = 3300

$\Rightarrow$ r = 22000

It is also given that the amounts of L used by the outlet to buy A and B are in the ratio 5:3.

$$\Rightarrow \frac{p * 95a}{q * 114a} = \frac{5}{3}$$

$\Rightarrow$ p = 2q

Also, the amounts of L the outlet received from the sales of A and B are in the ratio 5:9.

$\Rightarrow \dfrac{(p-800)*110a}{q*132a} = \dfrac{5}{9}$

$\Rightarrow \dfrac{(2q-800)*110a}{q*132a} = \dfrac{5}{9}$

$\Rightarrow$ q = 600

Therefore, p = 2q = 2*600 = 1200.

It is given that the outlet received 88000 units of L by selling A during the day.

$\Rightarrow$ (p-800)*110a = 88000

$\Rightarrow$ (1200-800)*110a = 88000

$\Rightarrow$ 44000a = 88000

$\Rightarrow$ a = 2

We can fill the entire table and answer all the questions.

				No. of units			
Currency	Buying rate	Base rate	Selling rate	Opening stock	Buy	Sell	Closing stock
A	190	200	220	2500	1200	400	3300
B	228	240	264	4800	600	600	4800
C	1.9	2	2.2	48000	22000	19000	51000

From the table we can see that the currency outlet bought 1200 units of A.

VIDEO SOLUTION

52. **C**

Let '100x' be the number of smartphones sold in year 2016.

Total revenue generated by Azra = 40x*15000 = Rs. 600000x

Profitability is defined as the profit as a percentage of the revenue. Therefore, profit generated by Azra = $\dfrac{10}{100}$ *600000x = Rs. 60000x

Total revenue generated by Bysi = 25x*20000 = Rs. 500000x

Profit generated by Bysi = $\dfrac{30}{100}$*500000x = Rs. 150000x

Total revenue generated by Cxqi = 15x*30000 = Rs. 450000x

Profit generated by Cxqi = $\dfrac{40}{100}$*450000x = Rs. 180000x

Total revenue generated by Dipq = 20x*25000 = Rs. 500000x

Profit generated by Dipq = $\dfrac{30}{100}$*500000x = Rs. 150000x

We can see that profit generated by Cxqi is the highest among all four brands. Hence, option C is the correct answer.

VIDEO SOLUTION

53. **A**

Let '100x' be the number of smartphones sold in year 2016.

Total revenue generated by Azra = 40x*15000 = Rs. 600000x

Profitability is defined as the profit as a percentage of the revenue. Therefore, profit generated by Azra = $\frac{10}{100}$ *600000x = Rs. 60000x

Total revenue generated by Bysi = 25x*20000 = Rs. 500000x

Profit generated by Bysi = $\frac{30}{100}$*500000x = Rs. 150000x

Total revenue generated by Cxqi = 15x*30000 = Rs. 450000x

Profit generated by Cxqi = $\frac{40}{100}$*450000x = Rs. 180000x

Total revenue generated by Dipq = 20x*25000 = Rs. 500000x

Profit generated by Dipq = $\frac{30}{100}$*500000x = Rs. 150000x

It is given that the market sales increased by 40%. Therefore, the number of smartphones sold in 2017 = 1.4*100x = 140x

It is given that Cxqi offered a 40% discount on its unit selling price in 2017 i.e. selling price in 2017 = 0.6*30000 = Rs. 18000

Also Cxqi's merket share increased by 15% whereas the other three brands lost 5% market share.

Brand	Maket Share (%)	No of units	Unit Selling Price (Rs.)	Profitibility(%)
Azra	35	49x	15,000	10
Bysi	20	28x	20,000	30
Cxqi	30	42x	18,000	20
Dipq	15	21x	25,000	30

Amount of profit generated by Azra = $\frac{10}{100}$*15000*49x = 73500x

Amount of profit generated by Bysi = $\frac{30}{100}$*20000*28x = 168000x

Amount of profit generated by Cxqi = $\frac{20}{100}$*18000*42x = 151200x

Amount of profit generated by Dipq = $\frac{30}{100}$*25000*21x = 157500x

We can see that profit of brands Azra, Bysi and Dipq increased in the year 2017 as compared to 2016. Hence, option A is the correct answer.

VIDEO SOLUTION

54. **B**

Let the total number of gold coins with the old woman be '9n'.
Total value of the assets with the old woman = 50 + 3*30 + 70 + 9n = 210+9n.
We know that the assets have been distributed in the ratio 1:2:3.
Therefore, Neeta must have received 35+1.5n (by value), Seeta must have received 70+3n and Geeta must have received 105+4.5n.
Further, it has been given that the gold coins distributed were in the ratio 2:3:4.
Therefore, the number of gold coins with Neeta must be '2n', Seeta must be '3n' and Geeta must be '4n'.
Seeta has '3n' gold coins. Therefore, the total value of the assets with her must be 70. Seeta could not have inherited all the flats. Therefore, Seeta must have received the house (worth 50 lakh) and 20 lakh from bank deposits.

We know that Geeta did not receive Rs. 30 lakh from the bank deposits. Therefore, Neeta must have received Rs. 30 lakh.
The remaining 5 lakh must be contributed by the gold coins (Since there is no other asset worth 5 lakh).
=> 5 + 1.5n = 2n

=> 0.5n = 5
=> n = 10
The old-woman must have had 10*9 = 90 gold coins. Therefore, option B is the right answer.

VIDEO SOLUTION